

Practice Problem Set 2

1. What type of tie is being described? Is it a symmetric, asymmetric, or anti-symmetric tie? If it's asymmetric, is it described as being reciprocal? If it is not clearly reciprocal, assume it is not. Is the connection type based on a flow, interaction, social relation, or similarity? The relationship to consider is underlined (10 pts)

	Symmetry	Reciprocal?	Type
1) Smriti is <u>classmates</u> with Emma.	_____	_____	_____
2) Youssef is your <u>roommate</u> . (You don't interact really)	_____	_____	_____
3) Michelle is the <u>parent</u> of Miranda.	_____	_____	_____
4) You <u>stan</u> (send attention to) Beyonce.	_____	_____	_____
5) Tom and Tom's boss <u>hate</u> each other. (Send hatred)	_____	_____	_____
6) Said and Lu <u>work together</u> .	_____	_____	_____
7) Alex has a <u>crush</u> (sends sentiment) on Sasha.	_____	_____	_____
8) Russell is <u>married</u> to Ciara.	_____	_____	_____
9) Jeff and Jeff's child both <u>like pizza</u> .	_____	_____	_____
10) Ali is <u>sisters</u> with AJ.	_____	_____	_____

2. Is the described tie Strong or Weak? Identify at least two elements of time, frequency, intimacy, or exchange are helping to drive your decision. (6 pts)

- 1) You've had the same roommate for 3 years; you like each other and exchange advice often.
- 2) Your parent is a little annoying and you don't like talking to them on the phone, but they're alright.
- 3) You start a study group with some other students in your section for this class, and Ashley gives you class notes because you missed class.

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3. 1. In the space below, create a friendship graph (stick and line) with the provided friendship ties (8 points). You can just use the first initial, they all have different initials to start their names)

A nominates B as a friend.
B nominates A as a friend.
D nominates E as a friend.
D nominates H as a friend.
C nominates B as a friend.
H nominates K as a friend.
K nominates H as a friend.
I nominates K as a friend.

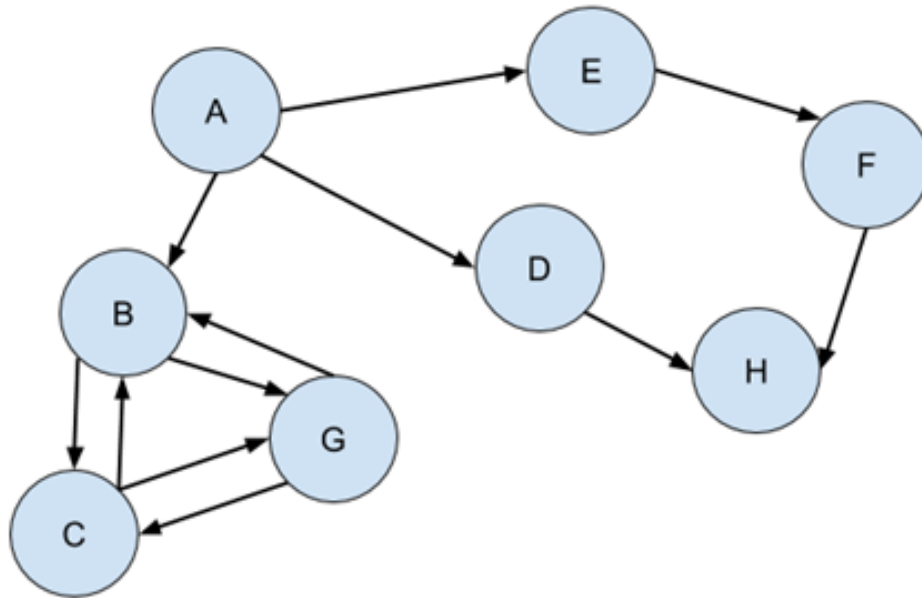
F nominates D as a friend.
F nominates E as a friend.
E nominates F as a friend.
E nominates D as a friend.
G nominates H as a friend.
I nominates J as a friend.
J nominates K as a friend.

Additional Questions

How many nodes in the graph?	_____ (1 pt)
How many edges in the graph?	_____ (1 pt)
Name two nodes that reciprocate the friendship.	_____ (1 pt)
Name two nodes that are a null dyad.	_____ (1 pt)
Name two nodes where the friendship is not reciprocated.	_____ (1 pt)

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4. Create a matrix reflecting the below graph. (11pts)



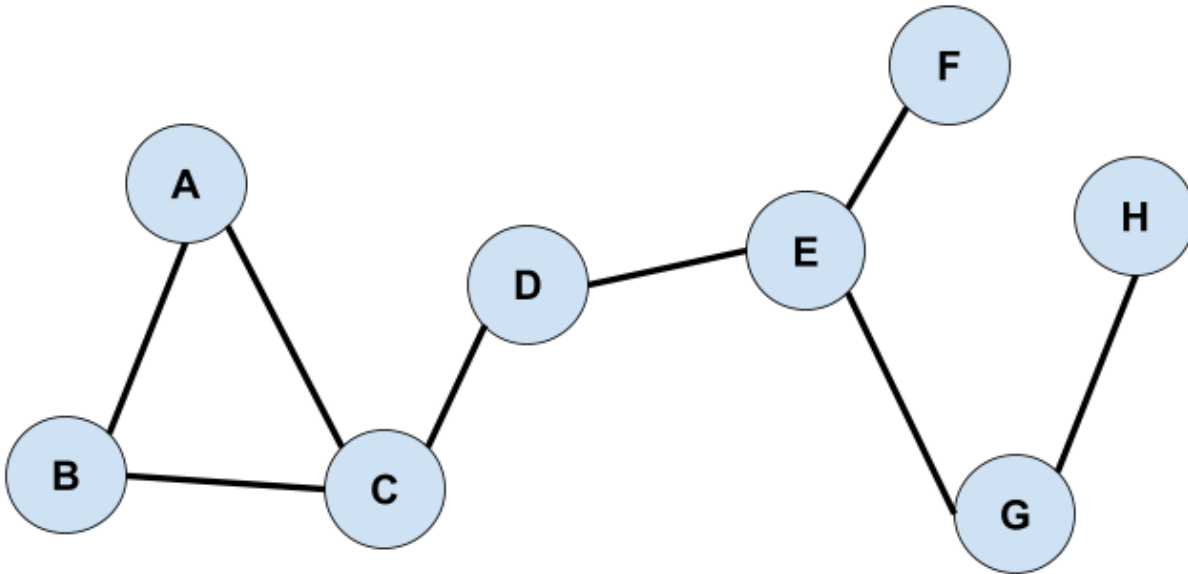
Additional Questions

- What type of graph is the above graph?
- What type of matrix is the above matrix?
- How many edges are there in graph/matrix?
- What is the shortest path from A to H?
- What is the shortest path from B to H?

_____ (1 pt)
_____ (1 pt)
_____ (1 pt)
_____ (1 pt)
_____ (1 pt)

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5. The following links are strong ties. What weak ties might we expect to exist according to the idea of Granovetter Transitivity? You can just draw dashed lines in for the weak ties that should exist. The number of points is a hint. (7pts)



6. Is the triad balanced or unbalanced from the perspective of the first named individual? (7pts)

- a. Ricardo likes Sasha and Terry, but Sasha doesn't like Terry.
- b. Alex hates Michelle and Emma. Michelle likes Emma.
- c. Sherlock likes Watson but doesn't like Moriarty. Watson doesn't like Moriarty.
- d. Alex hates Michelle and also Ali. Ali doesn't like Michelle.
- e. Lu likes Smriti and Beverly. Smriti also likes Beverly.
- f. Tom hates his boss. Tom likes his co-worker. His co-worker also hates his boss.
- g. Youssef likes Jeff and doesn't like Ali. Jeff likes Ali.

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7. Create a REACHABILITY matrix reflecting the below graph. (11pts)

